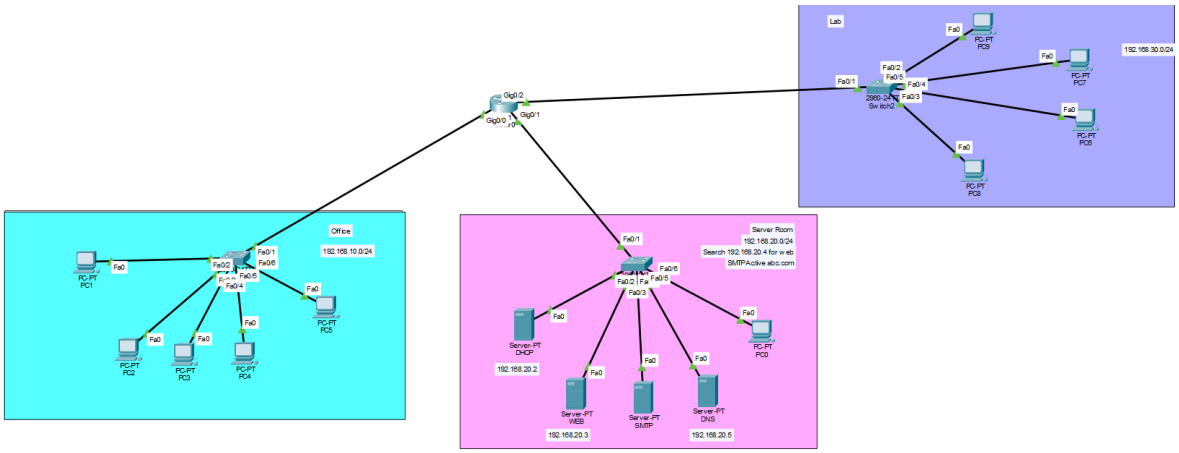
DHCP, DNS, Web and SMTP Server Setup in Packet Tracer

1. Introduction

In this project, we built a small office network in Cisco Packet Tracer. All the servers are kept in one Server Room. The goal is simple:

- Every PC gets its IP address automatically from a DHCP server.
- A DNS server turns names like `www.abc.com` into IP addresses.
- A web server shows a web page to all PCs.
- An SMTP (email) server lets users send and receive emails.



2. Network Design

The network has three parts (LANs). One router connects them. Each part has its own switch.

Place	Devices	Router Port	Network	Gateway
Office	PC1 to PC5	Gig0/0	192.168.10.0/24	192.168.10.1
Server Room	DHCP, WEB, SMTP, DNS, PC0	Gig0/1	192.168.20.0/24	192.168.20.1
Lab	PC6 to PC9	Gig0/2	192.168.30.0/24	192.168.30.1

The servers use fixed (static) IP addresses. Set them in Desktop > IP Configuration > Static.

Server	IP Address	Mask	Gateway	DNS
DHCP	192.168.20.2	255.255.255.0	192.168.20.1	192.168.20.5
WEB	192.168.20.3	255.255.255.0	192.168.20.1	192.168.20.5
SMTP	192.168.20.4	255.255.255.0	192.168.20.1	192.168.20.5
DNS	192.168.20.5	255.255.255.0	192.168.20.1	192.168.20.5

The domain name used in this project is `abc.com`.

3. Router Configuration

The router gives each LAN its gateway. The `ip helper-address` command sends DHCP requests from the Office and Lab to the DHCP server in the Server Room. Without it, those PCs will not get an IP address.

```
enable
configure terminal
hostname R1

interface gigabitEthernet0/0
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.20.2
 no shutdown
 exit

interface gigabitEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit

interface gigabitEthernet0/2
 ip address 192.168.30.1 255.255.255.0
 ip helper-address 192.168.20.2
 no shutdown
```

```
 exit

end
write memory
```

4. DHCP Server Setup

Open DHCP Server > Services > DHCP. Turn the service **On**. Add three pools and click **Save** after each one.

Pool Name	Default Gateway	DNS Server	Start IP	Mask	Max Users
Office_Pool	192.168.10.1	192.168.20.5	192.168.10.10	255.255.255.0	100
Lab_Pool	192.168.30.1	192.168.20.5	192.168.30.10	255.255.255.0	100
Server_Pool	192.168.20.1	192.168.20.5	192.168.20.10	255.255.255.0	100

The Server_Pool starts from .10 so it does not clash with the fixed server IPs (.2 to .5).

Then, on every PC, go to Desktop > IP Configuration and select **DHCP**.

5. DNS Server Setup

Open DNS Server > Services > DNS. Turn the service **On**. Add these records and click **Add** after each one.

Name	Type	Address
www.abc.com	A Record	192.168.20.3
mail.abc.com	A Record	192.168.20.4

6. Web Server Setup

Open WEB Server > Services > HTTP. Turn HTTP **On**. Edit `index.html`, write a simple message, and click Save.

To test, open the Web Browser on any PC and type `www.abc.com`.

7. SMTP (Email) Server Setup

The SMTP server has no CLI commands in Packet Tracer. It is set up from the GUI.

Open SMTP Server > Services > EMAIL:

- SMTP Service: **On**
- POP3 Service: **On**
- Domain Name: `abc.com`, then click **Set**
- Add users: `user1` with password `1234`, and `user2` with password `1234`

Now set up email on the PCs. Open Desktop > Email > Configure Mail.

Field	PC1	PC6
Your Name	user1	user2
Email Address	user1@abc.com	user2@abc.com
Incoming Mail Server	mail.abc.com	mail.abc.com
Outgoing Mail Server	mail.abc.com	mail.abc.com
User Name	user1	user2
Password	1234	1234

8. Testing

On PC1, open Command Prompt and run:

```
ipconfig /renew
ipconfig /all
ping 192.168.20.2
ping 192.168.20.4
ping 192.168.20.5
nslookup mail.abc.com
ping mail.abc.com
```

Then test email:

1. On PC1, open Email > Compose. Send a mail to `user2@abc.com`.
2. On PC6, open Email > Receive. The mail should arrive.

Results: PCs got IPs from DHCP, names resolved through DNS, the web page opened, and the email was delivered.

9. How to Handle Troubleshooting

The problem we faced: The Email app showed "cannot find the outgoing email server".

How we found the cause:

- `ping 192.168.20.4` worked. So the network and router were fine.
- `ping mail.abc.com` failed. So the name was not being resolved.
- `nslookup mail.abc.com` said "Non-existent domain". This means the DNS server did not have the record.

The fix: We added the A record `mail.abc.com` with address `192.168.20.4` on the DNS server and clicked **Add**. After that, email worked.

Other common problems:

- PC gets an IP like `169.254.x.x`: DHCP is not working. Check that the DHCP service is On and `ip helper-address` is set on the router.
- DNS server not shown: use `ipconfig /all`. The simple `ipconfig` does not show it.
- Name does not resolve: check that DNS service is On and the record spelling is exact.
- Router port is down: use `no shutdown`, check the cable, and run `show ip interface brief`.
- Quick email fix: write `192.168.20.4` instead of the name in the Incoming and Outgoing Mail Server fields.

10. Conclusion

The network works as planned. All PCs in the Office, Lab and Server Room get their IP address from the DHCP server. The DNS server resolves `www.abc.com` and `mail.abc.com`. The web page opens on all PCs, and users can send and receive email through the SMTP server. The one error we faced was a missing DNS record, and it was fixed easily using ping and nslookup.